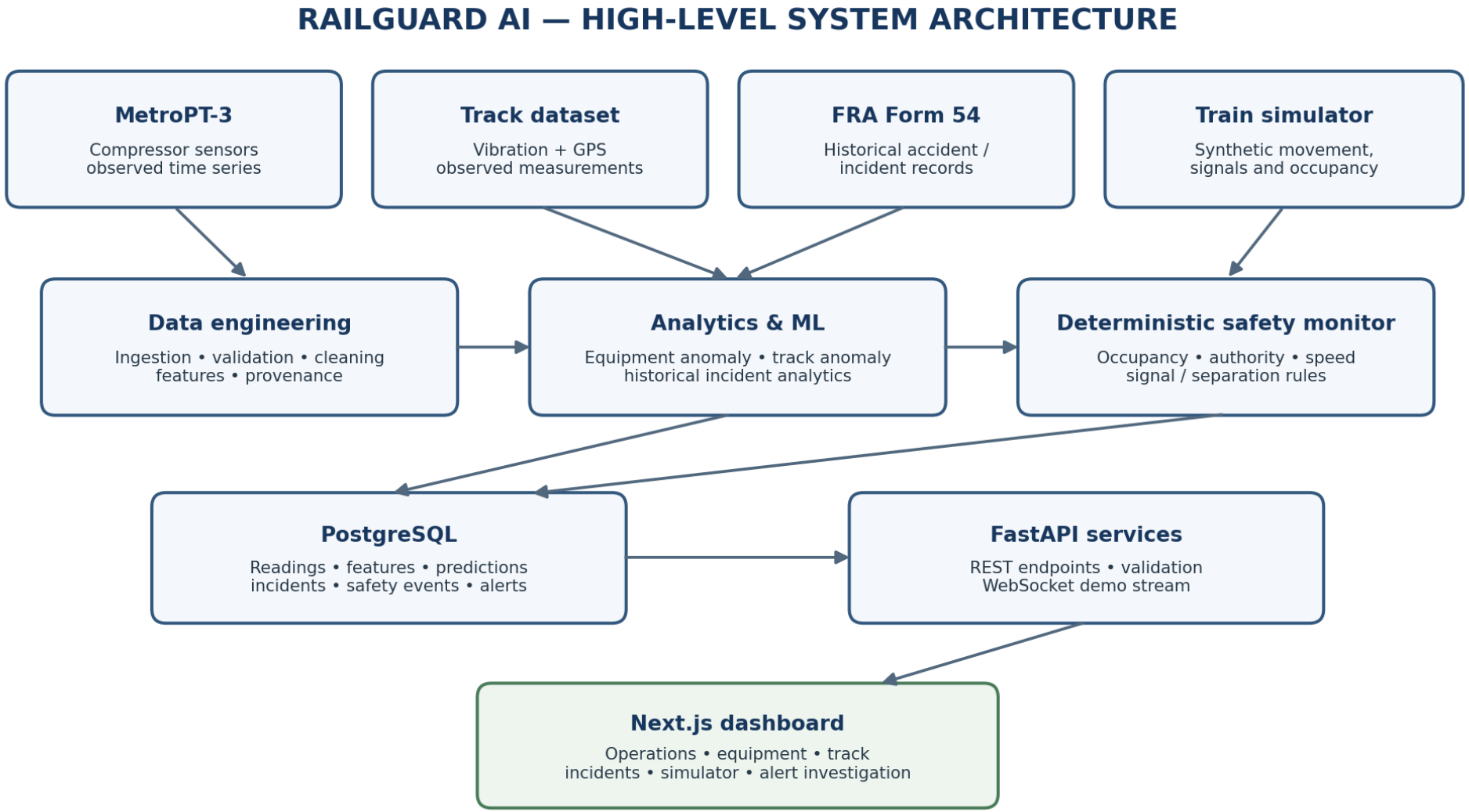
RAILGUARD AI — SYSTEM DIAGRAMS

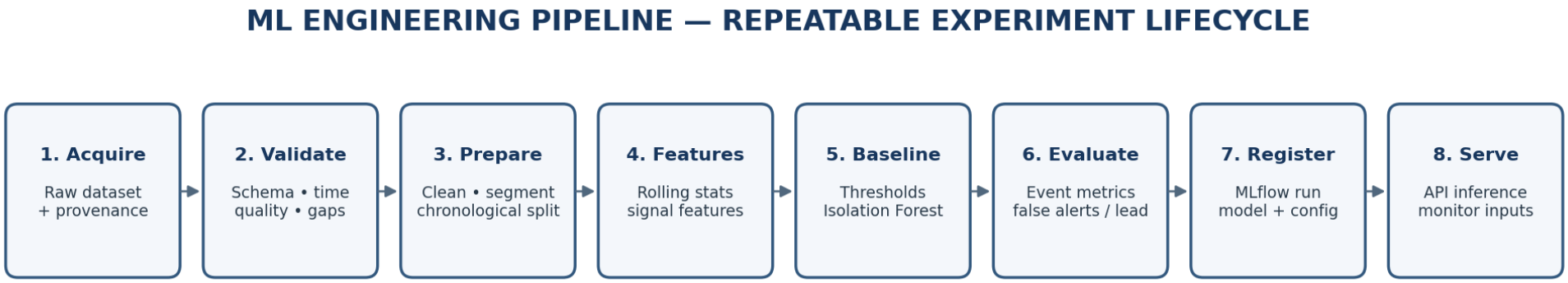
Conceptual diagrams for the project master specification • Version 1.0

A1. High-level system architecture



Observed public datasets and synthetic simulator feed separate processing paths. ML services and deterministic safety monitoring remain distinct before results are persisted and presented.

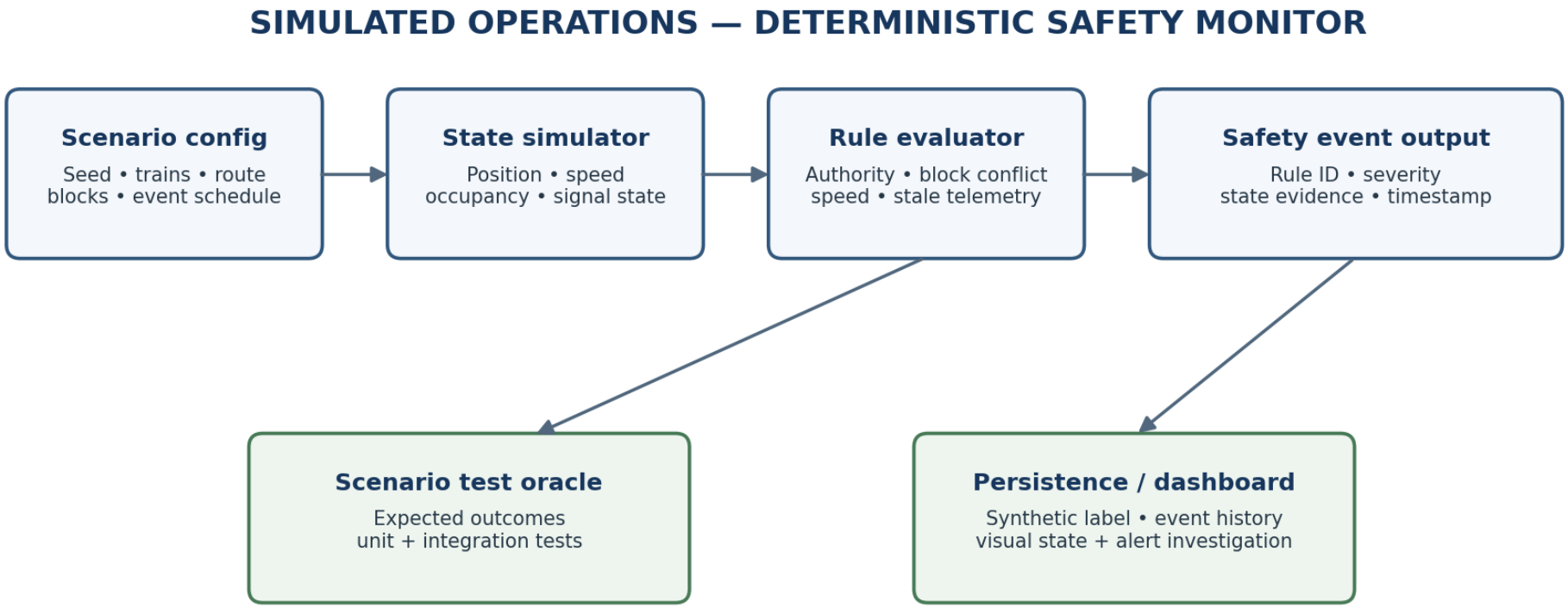
A2. ML engineering lifecycle



Keep raw data immutable • record dataset checksum / code commit • prevent future-data leakage

This is the repeatable workflow for each ML task. Use chronological / event-aware splits and preserve dataset, feature and code provenance.

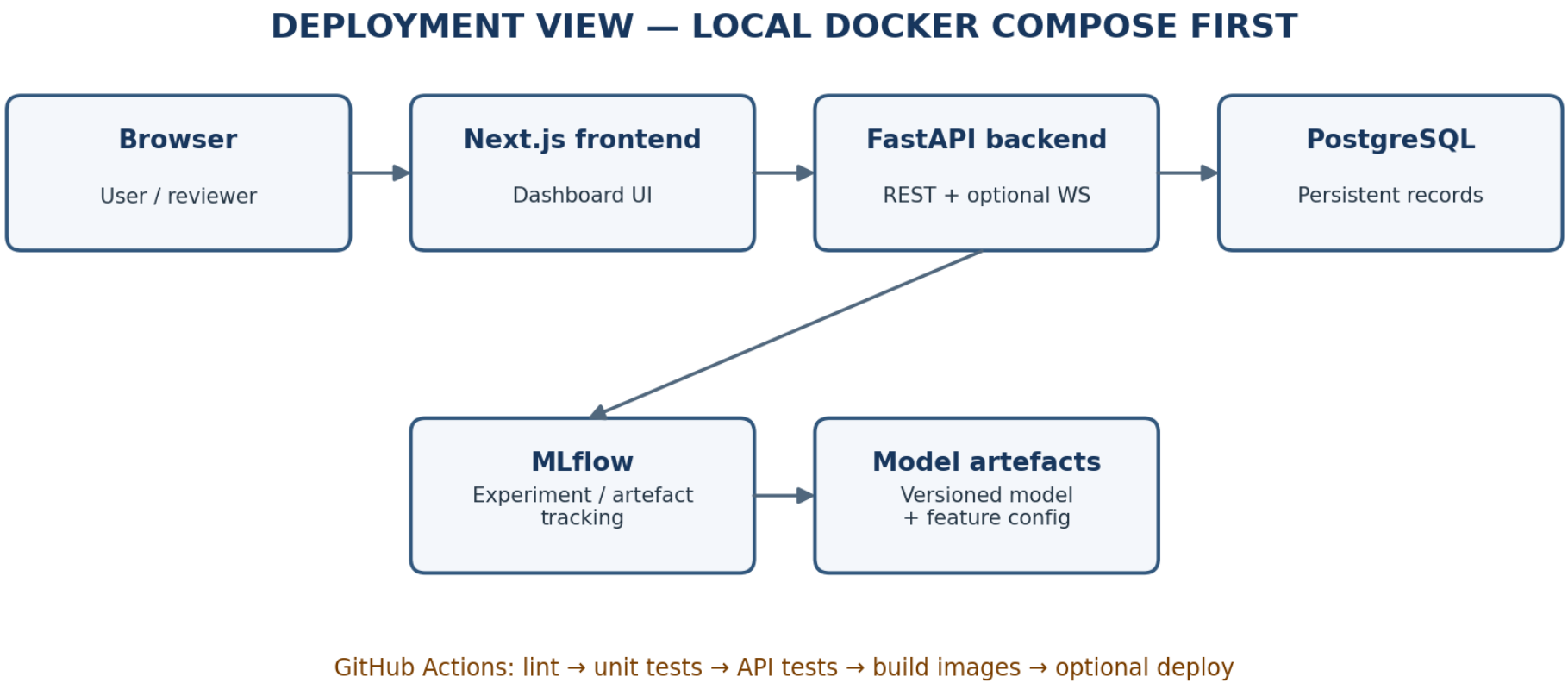
A3. Simulator and rule-engine flow



ML anomaly scores are supplementary; they cannot suppress or override a rule-engine event.

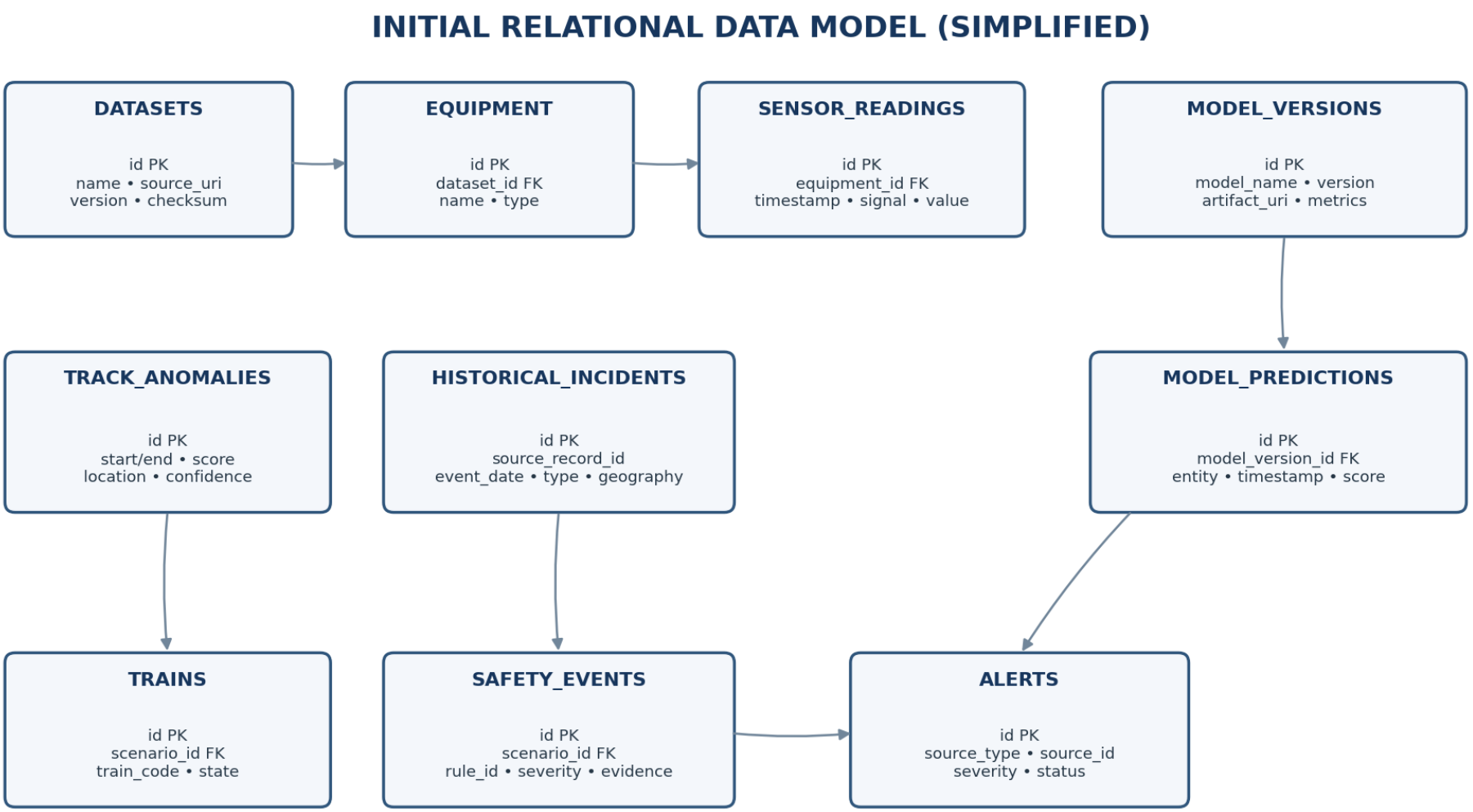
Scenario generation, state simulation, deterministic rule evaluation and expected-outcome testing are separate components. ML does not override rules.

A4. Deployment view



Start with a local Docker Compose environment. Add hosted deployment only after the API, database, frontend and model artefacts work reproducibly.

A5. Initial database entity relationship view



FK links shown conceptually; add scenario, route, track-section and audit tables as implementation requires.

This is a starting logical data model, not a final migration specification. Refine keys, constraints, indexes and cardinalities during implementation.